Akoizo Original Practice - Form 5

Division B | Ping Pong Parachute | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) Quadrupling the canopy area changes terminal velocity by a factor of:

A. 1/2

B. 1/4

C. 2

D. 1/16

Answer: _____

2. (2 points) For stable flight, the center of gravity should be:

Unit: Building a legal, stable rocket.
Worked-case data: A rocket's CG is 18 cm from the nose and its CP is 21 cm from the nose. The bottle diameter is 8 cm. Is it stable enough?
Separate investigation observations:
Flight 1: Rocket corkscrews and lands near a wall.
Inspection: One fin is twisted about 8° from the body line.
Flight 2: Fin realigned: rocket flies straight up.

A. Ahead of the center of pressure

B. Behind the center of pressure

C. At the same point as the center of pressure

D. At the nozzle

Answer: _____

3. (2 points) A team has an incomplete practice log and a clean 16.00 s launch. What is its score?

A. 13.60

B. 16.00

C. 11.20

D. 4.00

Answer: _____

4. (2 points) Why is release near apogee ideal?

Unit: Deployment and separation.
Worked-case data: Video shows a payload leaving the rocket 0.3 s after launch, well before apogee, and the parachute tangling.
What should change?
Separate investigation observations:
Cup depth 2 cm: Payload left during climb in 4 of 5 flights.
Cup depth 4 cm: Payload stayed attached in 3 of 5 flights.
Cup depth 3 cm: Payload released near apogee in 5 of 5 flights.

A. It is required by the rules

B. It avoids the 0.85 multiplier

C. The parachute begins its descent from the greatest height at low speed

D. It makes the rocket fly higher

Answer: _____

5. (4 points) Calculate apogee for an observer 12 m away sighting 40° (tan 40° about 0.84), eye height 1.6 m.

Answer: _____
_____
_____

6. (4 points) Why does a very light rocket not always fly higher?

Unit: How an air-powered bottle rocket climbs.
Worked-case data: A rocket reaches apogee 1.2 s after launch, and the payload then descends for 11.5 s. What share of time aloft came from the descent?
Separate investigation observations:
40 psi: Peak about 12 ft; time aloft 8.1 s.
55 psi: Peak about 18 ft; time aloft 12.4 s.
70 psi: Peak about 24 ft; time aloft 16.0 s in a 30 ft hall.

Answer: _____
_____
_____

7. (4 points) Write a one-sentence pressure justification for a 22-foot ceiling, using example data.

Answer: _____
_____
_____

8. (4 points) Why does adding tape to the payload matter?

Unit: Parachute physics: drag and slow descent.
Worked-case data: A 4.7 g payload uses a 40 cm diameter dome (C_d about 1.5). Estimate its descent speed and the time to fall 7 m.
Separate investigation observations:
30 cm canopy: Opens every time; 7.8 s descent.
60 cm canopy: Opens 6 of 10 times; 14.9 s when open, 3 s when not.
50 cm canopy, looser fold: Opens 10 of 10; 12.6 s descent.

Answer: _____
_____
_____

9. (6 points) Solve this case and explain your reasoning: A rocket's CG is 18 cm from the nose and its CP is 21 cm from the nose. The bottle diameter is 8 cm. Is it stable enough?

Answer: _____
_____
_____

10. (6 points) Solve this case and explain your reasoning: A team with a complete log records 14.20 s on launch 1 (clean) and 19.80 s on launch 2, but launch 2 grazed a light on the way up. What is its score?

Answer: _____
_____
_____

11. (6 points) Solve this case and explain your reasoning: Video shows a payload leaving the rocket 0.3 s after launch, well before apogee, and the parachute tangling. What should change?

Answer: _____
_____
_____

12. (2 points) Which launch pressure is best justified?

Supplied synthetic observations:
Chart: 45 psi -> 15 ft, 55 psi -> 19 ft, 65 psi -> 22 ft.
Spread: About ±1 ft at each pressure.
Venue: Ceiling announced at 21 ft.

A. About 50 psi

B. 65 psi

C. 45 psi

Answer: _____

13. (2 points) Define “Thrust” in the context of Ping-Pong Parachute.

Answer: _____
_____
_____

14. (2 points) Define “Required parameters” in the context of Ping-Pong Parachute.

Answer: _____
_____
_____

15. (2 points) Define “Drag coefficient (C_d)” in the context of Ping-Pong Parachute.

Answer: _____
_____
_____

16. (2 points) Define “Ceiling height” in the context of Ping-Pong Parachute.

Answer: _____
_____
_____

17. (2 points) Define “Drag” in the context of Ping-Pong Parachute.

Answer: _____
_____
_____

18. (2 points) Define “Outlier” in the context of Ping-Pong Parachute.

Answer: _____
_____
_____